Moment Generating Function

$$M(t) = \begin{cases} \sum_x e^{tx} P(X = x) & X \text{ discrete} \\ \underbrace{\int_{-\infty}^{\infty} e^{tx} f(x)\, dx}_{\text{Laplace T-form of PDF } f(x)!} & X \text{ continuous} \end{cases}$$

$M(t)$ uniquely determines the cumulative probability distribution $P(X < x)$

Thm.: $M^{(n)}(0) = E(X^n)$

$M'(t)$: MEAN: $E(X) = \mu$

$M''(t)$: VARIANCE: $E(X^2) - (E(X))^2 = \sigma^2$

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$M^{n}(t)$: "MOMENTS": $E(X^n)$

Ex.: Poisson $X \sim \text{poisson}(\lambda)$

$$M(t) = \sum_k e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_k \frac{(e^t \lambda)^k}{k!} =$$

$$= e^{-\lambda} e^{\lambda e^t} = \underbrace{e^{\lambda(e^t - 1)}}_{M(t)}$$

Ex.: Normal $X \sim N(0,1)$

$$M(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-x^2/2} dx =$$

$$\left[\frac{x^2}{2} - tx = \frac{1}{2}(x^2 - 2tx) = \frac{1}{2}(x-t)^2 - \frac{1}{2} t^2 \right]$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2 + \frac{1}{2}t^2} dx =$$

$$= e^{\frac{1}{2}t^2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2} dx = \underbrace{e^{t^2/2}}_{M(t)}$$

Proof: For Poisson Distribution

$$M(t) = e^{\lambda(e^t - 1)}$$

$$M'(t) = e^{\lambda(e^t-1)} \cdot \lambda e^t$$

$$M''(t) = e^{\lambda(e^t-1)} (\lambda e^t)^2 + e^{\lambda(e^t-1)} \lambda e^t$$

$$E(X) = M'(0) = \lambda$$

$$E(X^2) = M''(0) = \lambda^2 + \lambda$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$